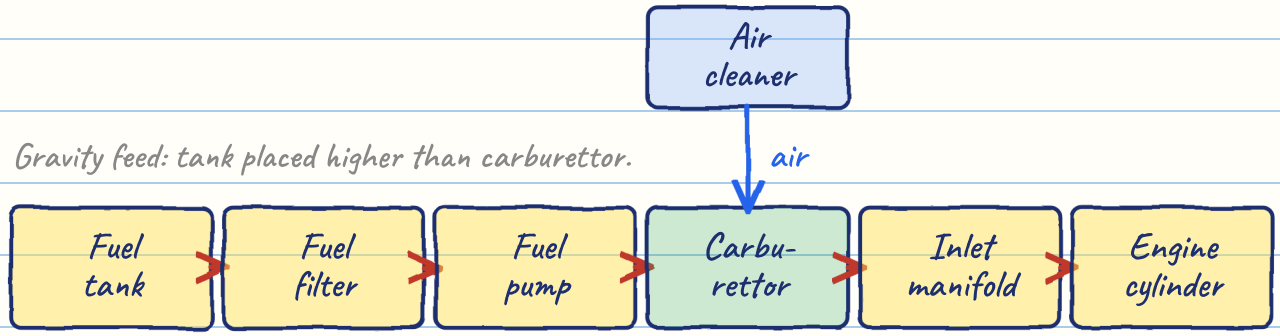
UNIT - II : FUEL, COOLING, LUBRICATION & IGNITION SYSTEMS

2.1 Fuel Supply System of an S.I. Engine

- Function : store the fuel, deliver it at the correct rate and prepare a combustible air-fuel mixture of correct strength for every speed & load.



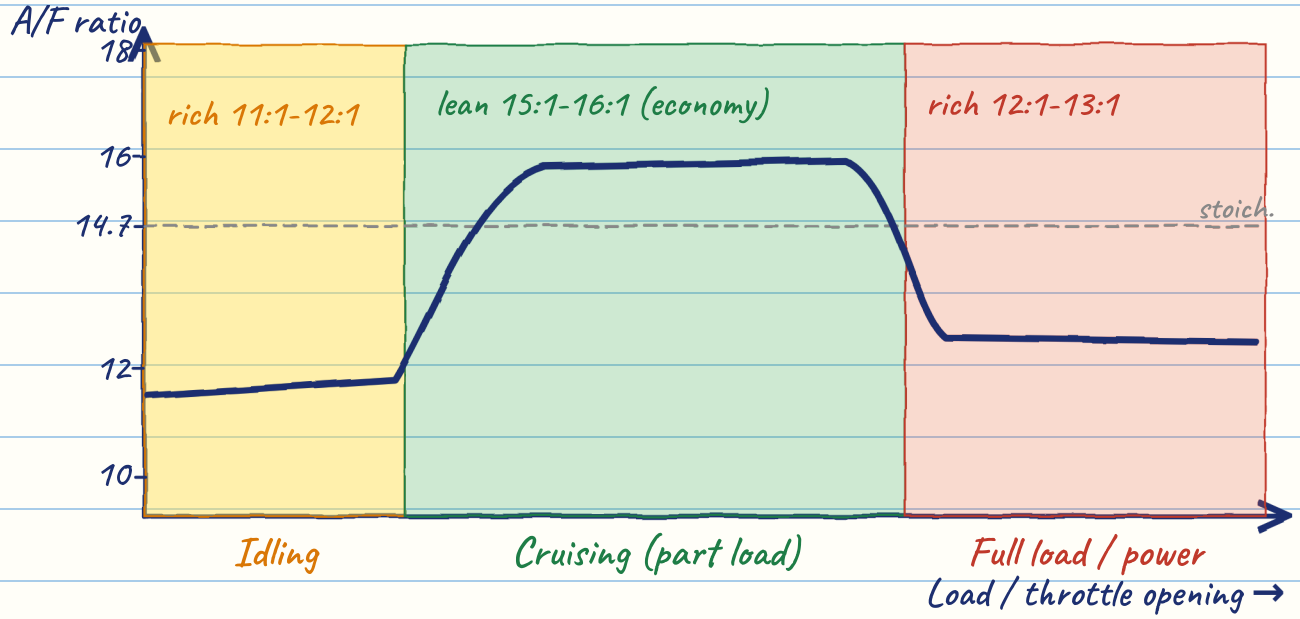
Fuel (petrol) is lifted from the tank by the pump, cleaned in the filter, metered & mixed with filtered air in the carburettor, and the mixture is drawn through the manifold into the cylinder.

- Fuel tank : sheet-steel / plastic, has filler cap with vent, baffles (stop surging) & a float type level gauge.
- Fuel lines : copper / nylon pipes carry fuel from tank → pump → carburettor.
- Strainer / filter : removes dirt, water & gum (paper, ceramic or gauze element).
- Fuel pump : mechanical (diaphragm) or electrical; needed where tank is below the carburettor.
- Carburettor : meters the fuel & atomises it in the air stream.

2.2 Air-Fuel Ratio Requirements

- Complete combustion needs about 14.7 : 1 (stoichiometric, chemically correct) - air : fuel by mass.

- **Idling** - rich (11:1 to 12:1) because residual exhaust gas dilutes the charge.
- **Cruising / part load** - lean (15:1 to 16:1) for economy.
- **Full load / acceleration** - rich (12:1 to 13:1) for maximum power.



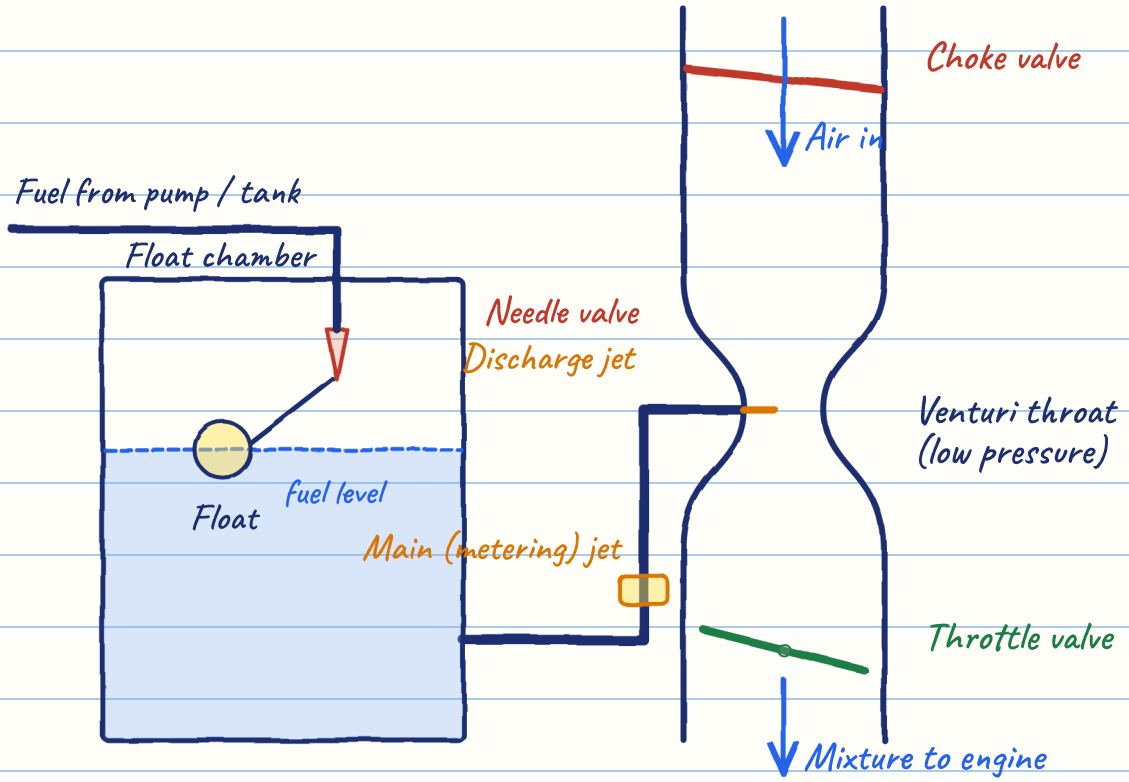
Rich mixture → max. power, more CO & HC. Lean mixture → best economy, higher NOx & cylinder temperature (risk of backfire).

2.3 Carburetion - Principle

- **Carburation** = process of preparing a combustible mixture of petrol & air in the proper ratio outside the cylinder.
- Based on the venturi principle (Bernoulli) : at the throat velocity is high ⇒ pressure falls (partial vacuum) ⇒ fuel is pushed from float chamber through the jet by the atmospheric pressure acting on the fuel surface.

$$\Delta p = \rho_a (C^2/2), \text{ fuel flow} \propto \sqrt{(\Delta p)}$$

2.4 Simple Carburettor



- **Working** : on the suction stroke air is drawn through the venturi; pressure at throat < pressure in float chamber ⇒ fuel is sprayed through the jet & vaporises. Throttle valve controls the quantity of mixture delivered to the engine ⇒ power.
- **Float chamber** keeps fuel level constant (≈ 4-5 mm below jet tip) to prevent fuel spilling when engine is stopped. The needle valve shuts when the float rises.
- **Choke valve** is closed for cold starting → creates strong vacuum → very rich mixture.

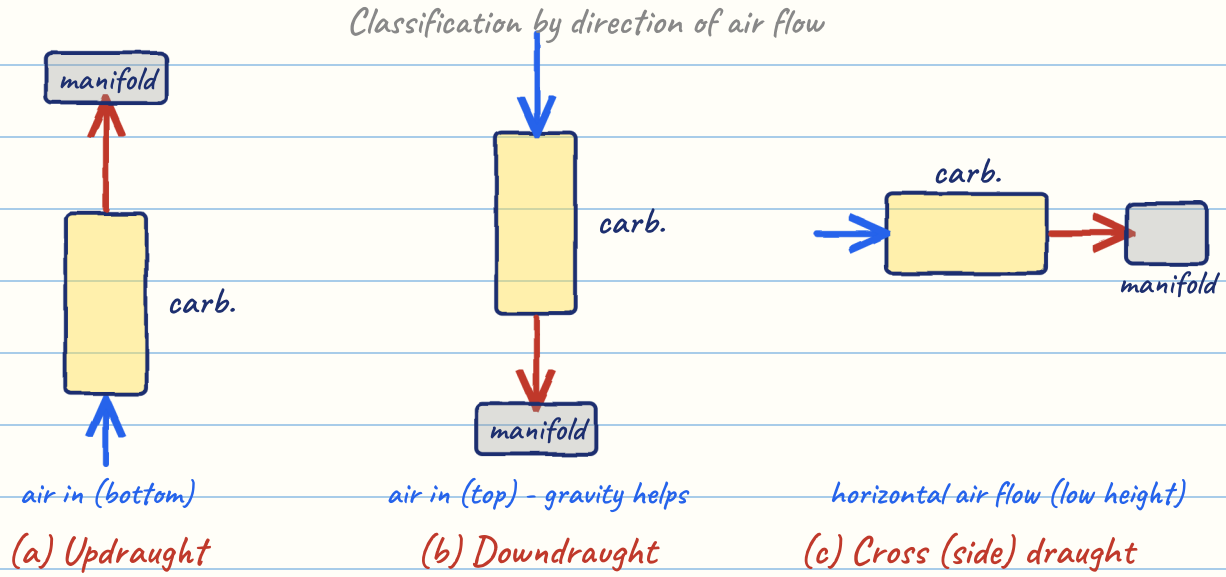
Defects of a simple carburettor

- Gives progressively **richer** mixture as throttle opens (fuel flow ∝ $\sqrt{\Delta p}$ but air flow density falls).
- Cannot give correct mixture for idling, cold start or acceleration, no economy at cruise.

2.5 Additional Devices in a Modern Carburettor

- **Compensating jet** - supplies air-bled fuel at high speeds to keep mixture from becoming rich.
- **Idling (slow-running) jet** - supplies rich mixture past the nearly closed throttle at low speed.
- **Accelerating pump** - a piston/diaphragm pump linked to the throttle which injects an extra shot of petrol for sudden acceleration.
- **Economiser** enriches at full load; **auto-choke** (bimetal) for cold start; **air bleed** improves atomisation.

2.6 Types of Carburettor

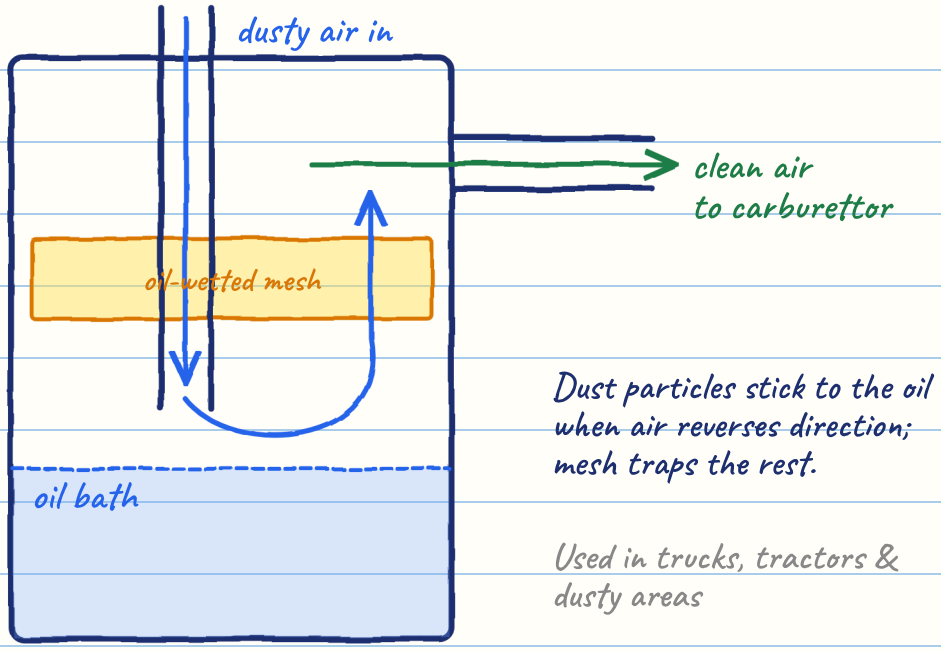


- (1) **By direction of airflow** : updraught, downdraught (most common - gravity helps fuel flow, easier cold start, better volumetric efficiency) and cross-draught.
- (2) **By venturi type** : **constant choke** (Solex, Zenith, Carter - area constant, depression varies) and **constant vacuum** (S.U., Constant Depression type - a piston moves to keep vacuum constant, variable choke).

- (3) Single, dual & multi-barrel types. **Solex** : downdraught constant-choke ; **S.U.** : horizontal variable-choke with tapered needle.

2.7 Air Filters (Air cleaners)

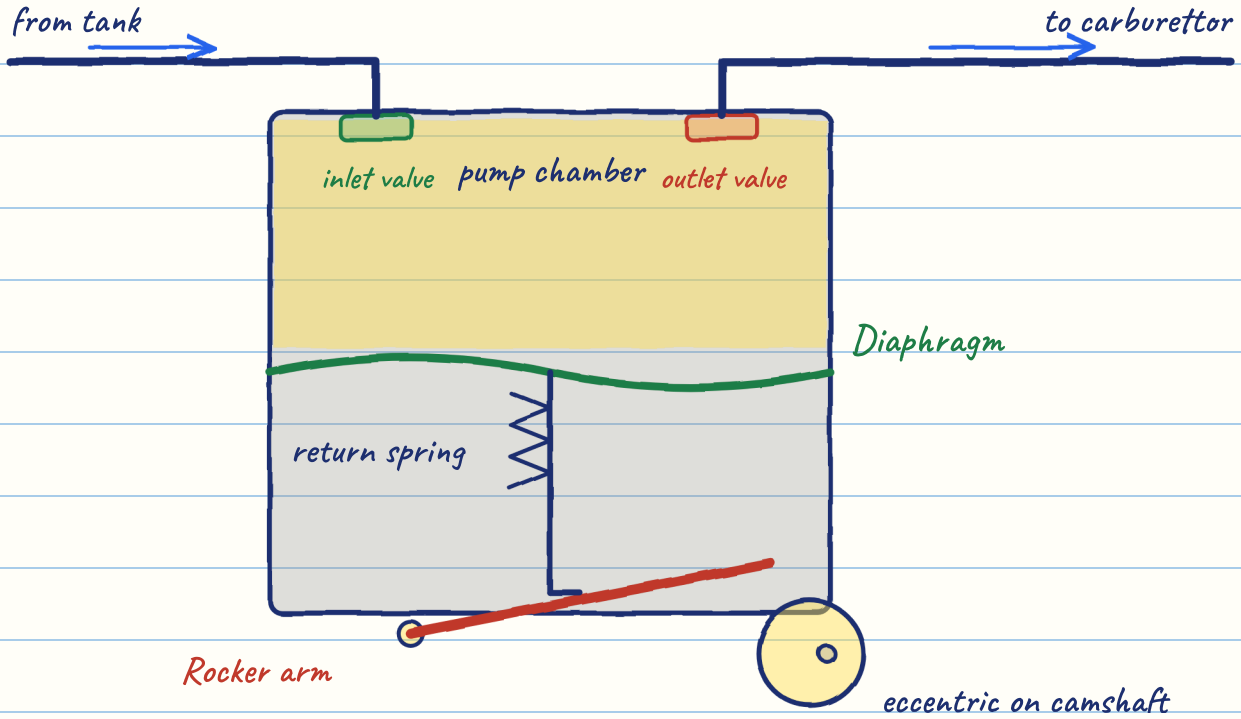
- Purpose : remove dust from the intake air (dust causes cylinder / ring wear) and silence intake noise.
- **Dry (paper) type** : pleated resin-impregnated paper element; cheap, efficient (99%), replaced periodically; most cars.
- **Oil-bath type** : air strikes oil surface, dust sticks to oil; wire mesh wetted with oil traps the rest; used in heavy duty / dusty places.
- **Viscous / centrifugal** types : oil-wetted mesh, cyclone pre-cleaners for trucks.



2.8 Fuel Pumps

- Needed when tank is below the carburettor (or fuel is injected under pressure).

(a) Mechanical (diaphragm) pump



- Driven by an eccentric on the camshaft through a rocker arm.
- **Suction stroke** : arm pulls diaphragm down against spring; vacuum opens inlet valve, fuel enters chamber from tank.
- **Delivery stroke** : when cam releases the arm the spring pushes the diaphragm up → inlet valve closes, outlet valve opens → fuel flows to carburettor. If the carburettor float needle is closed, the diaphragm stays down - pump idles.
- Delivery pressure ≈ 0.2 - 0.3 bar (low pressure). Simple, cheap, but vapour-lock problems.

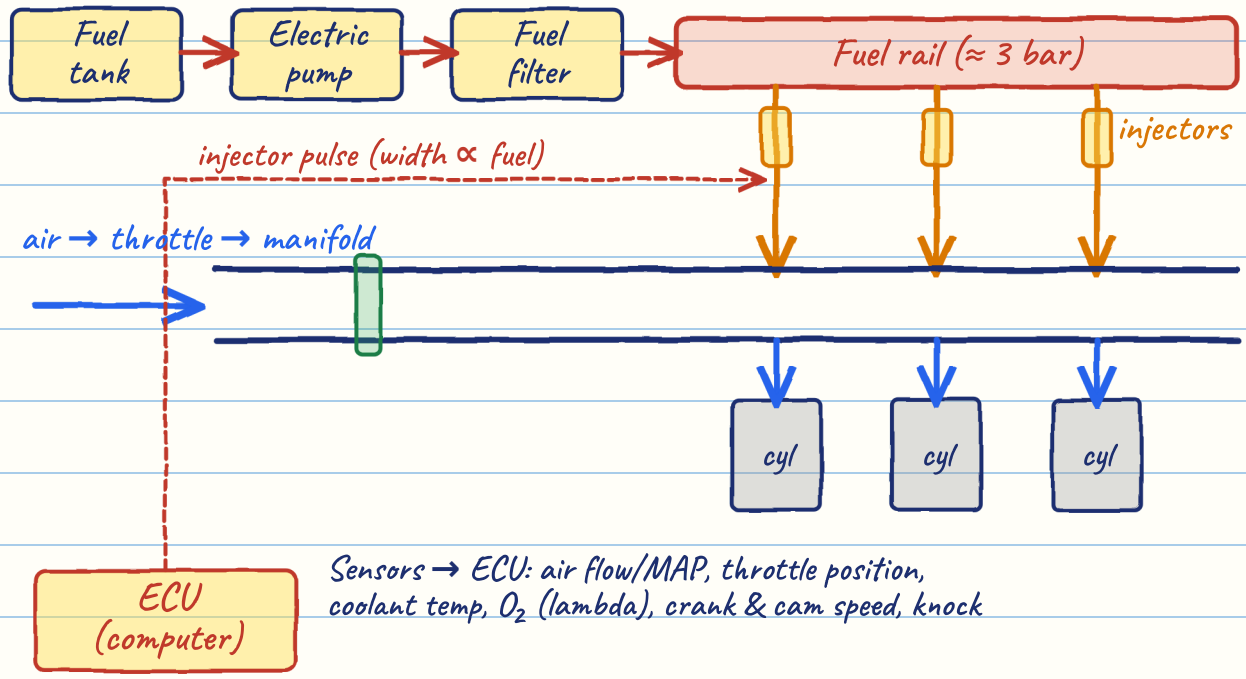
(b) Electrical fuel pump

- A **solenoid** pulls an armature/diaphragm; contact points open & close automatically (like a buzzer) giving reciprocating action. Fitted near the tank (pushes fuel ⇒ no vapour lock). In fuel-injected cars an **in-tank rotary (roller-cell / gerotor) pump** gives 3-5 bar.

- Advantages : delivers fuel even before the engine starts; can be placed anywhere; less vapour lock. Disadvantage : needs battery, noise, points burn.
- Fuel filter (2.9) : paper element (5-10 μm) / gauze between tank-pump-carburettor, replaced every 20-30 thousand km.

2.10 Gasoline (Petrol) Injection Systems

- Fuel is injected under pressure; computer meters it exactly. Advantages over carburettors : better mixture distribution, higher volumetric efficiency (no venturi restriction), precise A/F control for emission norms, better cold start & acceleration, ~ 10 % less consumption.



- Types : (a) Throttle body (single point) injection - TBI : 1 injector above the throttle. (b) Multi-point (port) injection - MPFI : injector for each cylinder at the inlet port; sequential or group firing. (c) Gasoline Direct Injection - GDI : injector sprays straight into the cylinder at 50-200 bar; stratified lean charge, ~ 15-20 % better economy.

- ECU reads sensors (MAF/MAP, TPS, coolant temp, lambda / O_2, crank speed) & varies injector pulse width. Closed loop with the O_2 sensor keeps $\lambda \approx 1$ for the 3-way catalytic converter.

2.11 Cooling System - Requirements

- About 30 - 35 % of the heat of fuel goes to the coolant & walls, 30-35 % to exhaust, 30 % to useful work.
- Gas temperature reaches 2000-2500 °C but metal & lubricating oil limits are low ⇒ cooling is must to : (1) avoid overheating & seizure of piston, (2) prevent oil film breakdown, (3) avoid pre-ignition / knocking, (4) keep valve & head strength, (5) maintain high volumetric efficiency.
- But over-cooling wastes heat, lowers thermal efficiency, causes fuel condensation & wear. Optimum cylinder wall temp. ≈ 80-95 °C (water outlet).

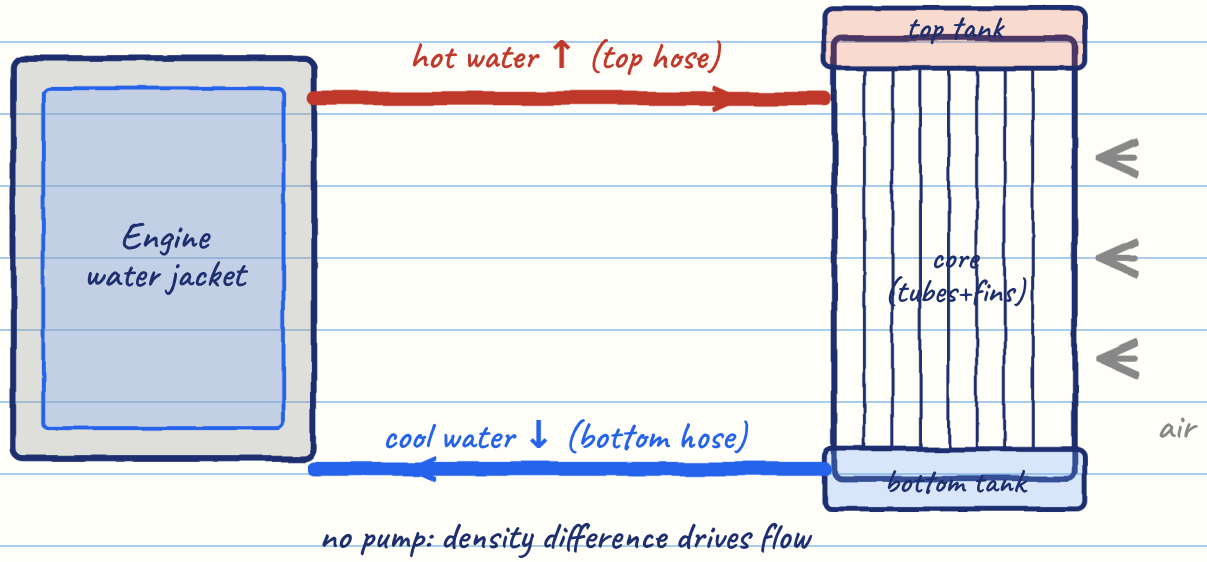
2.12 Air Cooling

- Heat is removed by air flowing over fins cast on the cylinder & head. Used in motorcycles, scooters, small engines & aircraft.
- Fins : greater surface, thin & closely spaced; fan / blower and cowling (baffles) are used in engines.
- Advantages : light, simple, no radiator / coolant leakage / freezing, quick warm-up. Disadvantages : non-uniform cooling, noisy, lower power density, not suitable for big engines.

2.13 Liquid (Water) Cooling

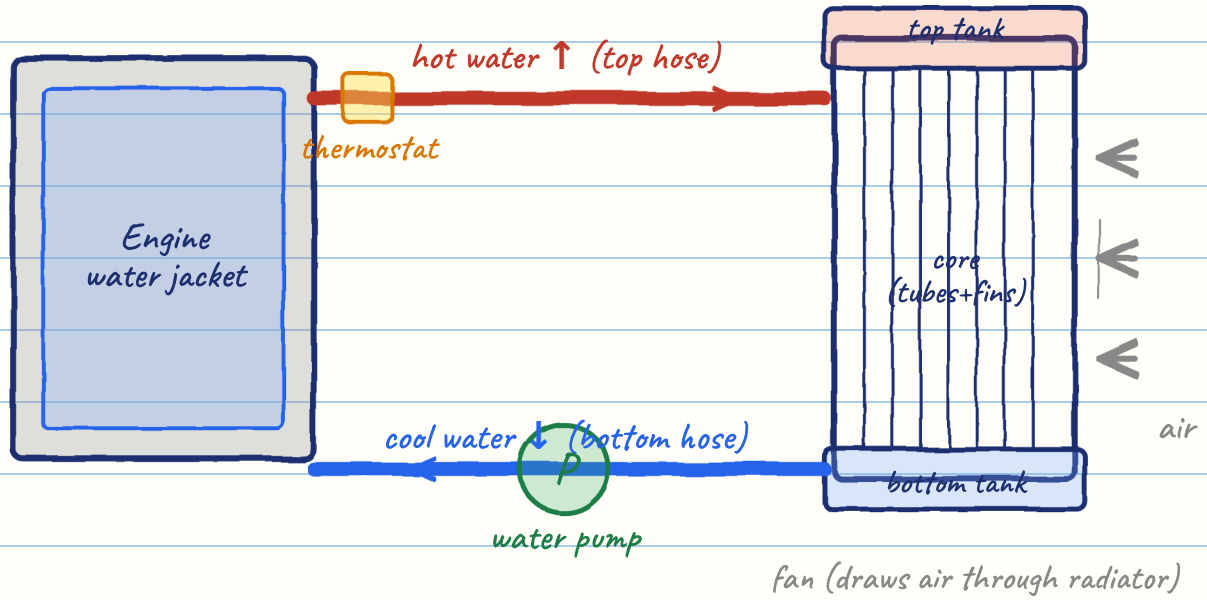
- Water jackets surround cylinders & head; heat taken by water → radiator → air. Coolant = water + ethylene glycol (anti-freeze) + corrosion inhibitor.

(a) Thermo-siphon system



- Water in the jacket heats, becomes lighter & **rises** to radiator top tank; cooled water is denser & **sinks** to bottom tank → back to engine. Flow due to density difference only.
- **No pump**, simple, but slow circulation, needs large water capacity & high radiator position ⇒ obsolete; used in old tractors.

(b) Forced circulation (pump) system



- **Water pump** (centrifugal, belt driven) circulates water at 20-60 l/min.
- **Thermostat** (wax or bellows type) stays closed until ≈ 82 °C - gives quick warm-up, then opens to the radiator; a bypass keeps flow in the block when closed.
- **Radiator** : top & bottom tanks joined by tubes with fins (cellular / tubular core), pressure cap (0.9-1 bar) raises boiling point. **Fan** (belt or electric) pulls air.
- Liquid cooling : uniform, quiet, higher power density (cars, buses) vs air cooling : light & simple (bikes).

2.14 Lubrication System

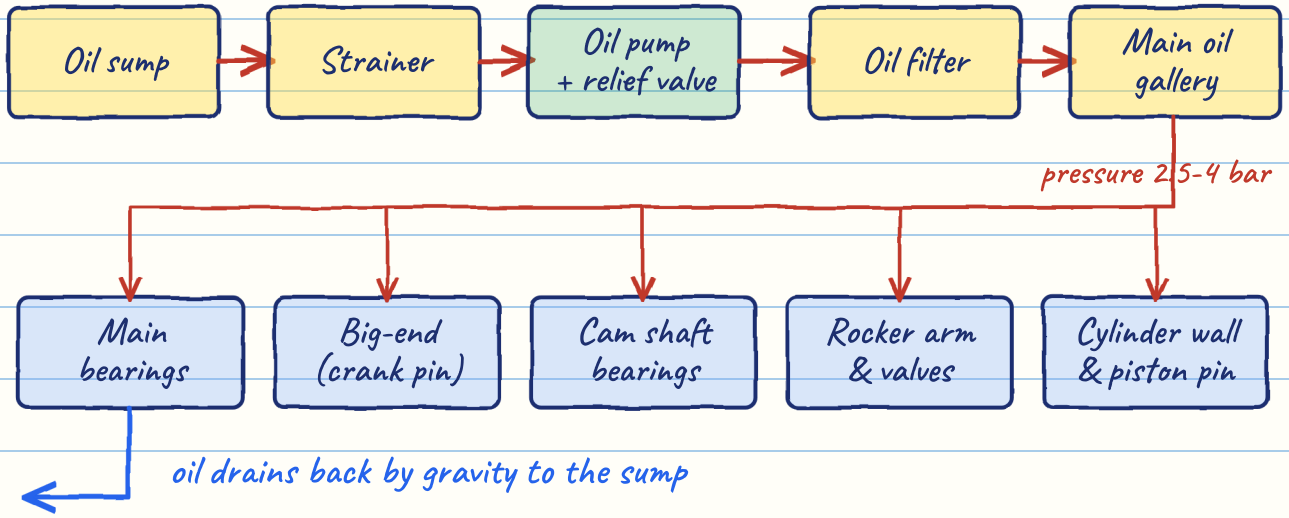
- **Functions of lubricant** : (1) reduce friction & wear, (2) cooling of piston & bearings, (3) cleaning (carries away wear particles), (4) sealing of piston rings, (5) cushioning shocks, (6) corrosion protection.
- Properties of lube oil : suitable **viscosity** & high viscosity index (VI), flash point, pour point, oxidation stability, detergent / dispersant additives, low volatility. SAE grades 10W-30, 20W-40.

Types of lubrication systems

(1) Splash / "Flash" system

- Connecting-rod **dippers** scoop oil from troughs in the sump / crankcase at each revolution & splash it on cylinder walls, piston pin, main & cam bearings. Used in small, low speed engines, lawn mowers. Oil level must be kept constant; variation in splash with speed.

(2) Pressurised (forced feed) system



- **Oil pump** (gear / rotor type) draws oil from sump through the strainer & forces it at 2.5-4 bar through the filter to the main gallery & drilled passages to bearings, cams, rockers; piston & cylinder wall are lubricated by oil thrown from the big-end. **Relief valve** limits the pressure.
- **Wet sump** : oil stored in the sump below the engine (cars). **Dry sump** : oil collected in a separate tank by a scavenge pump (racing, aircraft; better when tilted).
- Combination of splash + pressure also used (semi-pressure).

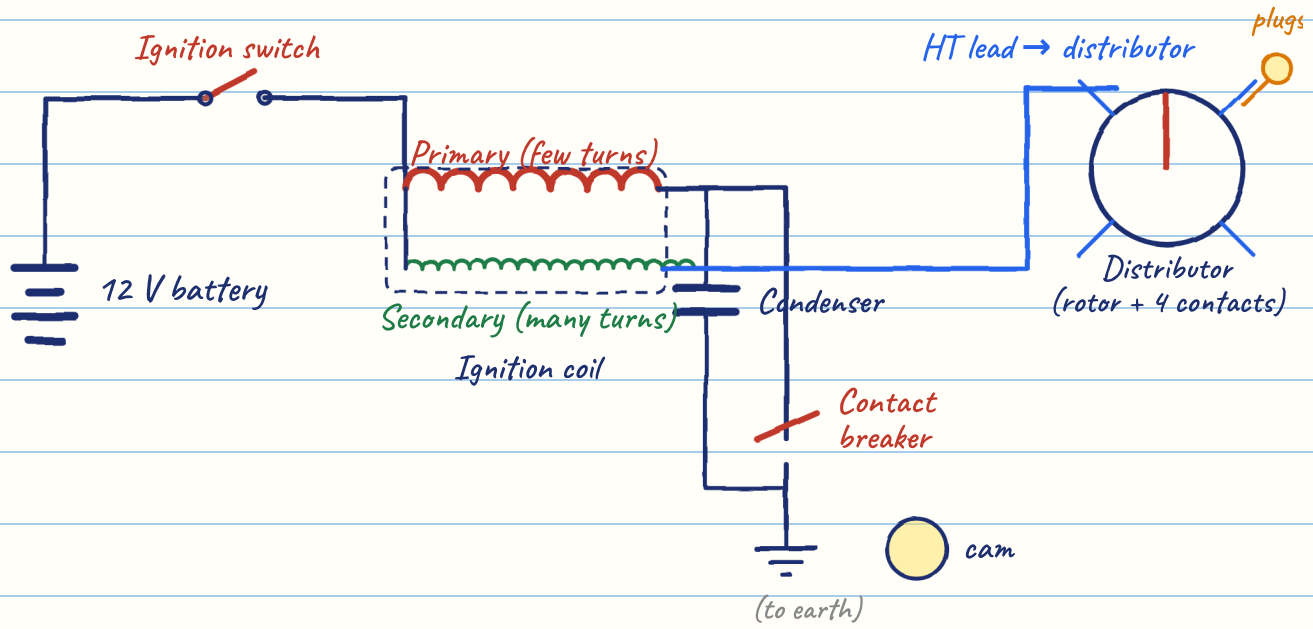
(3) Mist (petroil) lubrication

- Used in **two-stroke** engines : lube oil (2-6 %) is mixed with the petrol (or injected into intake). The charge mist passes through the crankcase; oil droplets deposit on parts, remaining oil is burnt with the fuel. Simple, light, **no sump**; but gives smoke & carbon deposits, poor control.

2.15 Ignition System

- **Function** : to supply a high-voltage spark (≈ 8,000 - 30,000 V) at the right instant (at the end of compression) to every cylinder in the **firing order** (4-cyl : 1-3-4-2 or 1-2-4-3).
- **Requirements** : (1) adequate spark energy & voltage, (2) correct timing with advance as per speed & load, (3) distribute in firing order, (4) reliable, durable, easy to maintain.

2.16 Battery (Coil) Ignition System



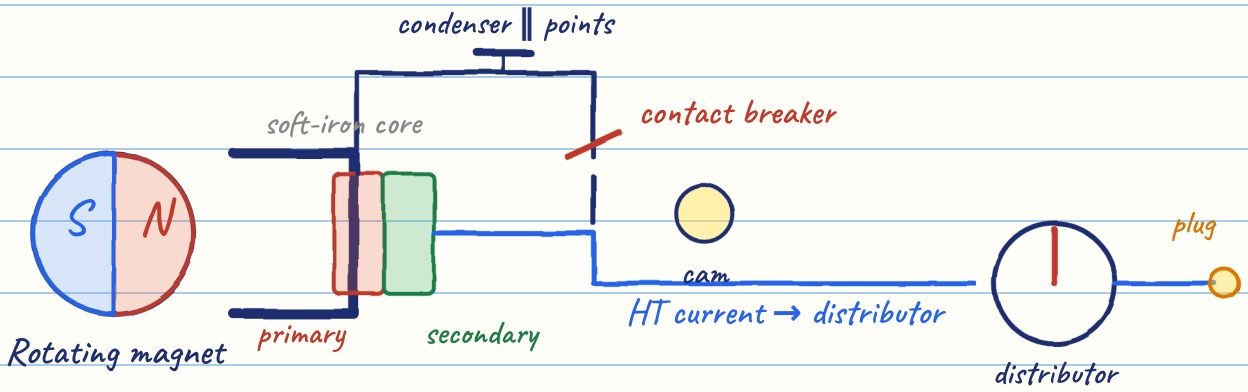
- Components : battery, ignition switch, **ignition coil** (auto-transformer), **contact breaker** & cam, **condenser**, **distributor**, spark plugs.
- **Working** : with points closed, current ≈ 3-4 A flows in the primary building a magnetic field. The cam opens the points → primary current **collapses suddenly** → flux collapses & cuts the many secondary turns ⇒ high voltage (≈ 20-30 kV) induced in the secondary → rotor → distributor contact → spark plug → spark jumps the gap (0.6-0.8 mm).

- **Condenser** (0.2 μF) across the points absorbs the self-induced primary emf (≈ 250 V) ⇒ prevents arcing at points & speeds up current collapse (higher secondary voltage).

$$V_2 = V_1 \times (N_2 / N_1) \text{ (turns ratio} \approx 100 : 1)$$

- A **ballast resistor** limits current at low speed & is bypassed at starting.
- Disadvantages : poor high speed spark (current has little time to build), points wear & need service; battery run-down effect at starting.

2.17 Magneto Ignition System



Rotating magnet cuts the coil → flux changes → emf in primary;
when the points open the primary current collapses → HV in secondary.

- A **permanent magnet** (rotating) or rotating armature generates the emf itself ⇒ **no battery**. It combines the generator, coil, breaker, condenser and distributor.
- **Types** : rotating armature, rotating magnet, polar inductor type. Used in motorcycles, scooters, aircraft & small engines (reliable at high speeds).

Point	Battery ignition	Magneto ignition
Battery	needed	not needed

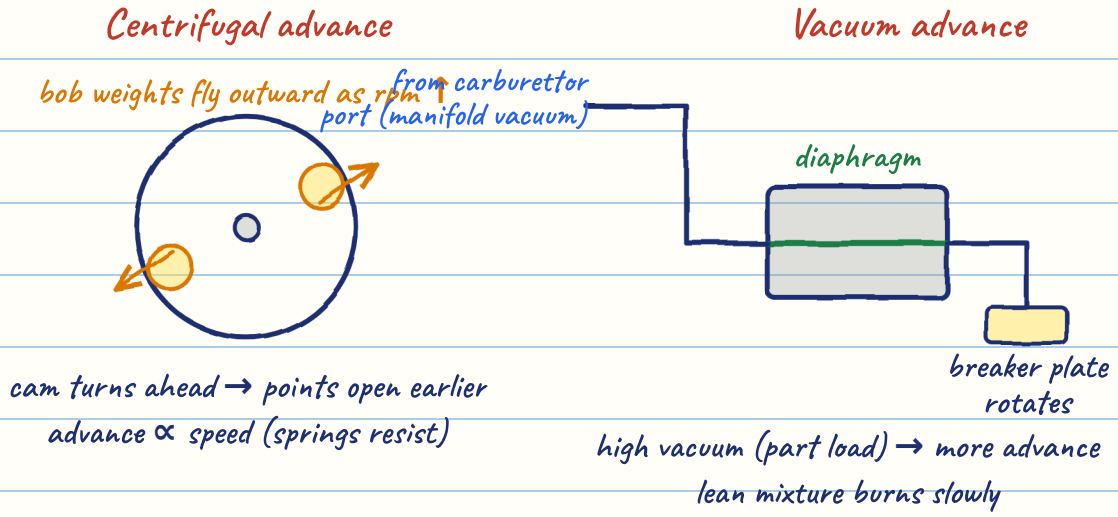
Point	Battery ignition	Magneto ignition
Starting	good spark at cranking	poor at low (cranking) speed
High speed	spark weak	very good
Maintenance	more (battery)	less
Use	cars	2-wheelers, aircraft

2.18 Electronic Ignition Systems

- **(a) Transistorised / contact-breaker assisted** : the old points carry only a small (≈ 0.5 A) current to switch a power transistor, which breaks the heavy primary current ⇒ points last longer, secondary voltage higher.
- **(b) Contact-less (triggered) type** : the breaker is replaced by a trigger device : **magnetic pick-up (reluctor)**, **Hall-effect sensor** or **optical** sensor on the distributor shaft; the **igniter / ECU module** switches primary current.
- **(c) Capacitor discharge (CDI)** : capacitor charged to ≈ 400 V, discharged through the primary by a thyristor (SCR) ⇒ very fast rise of secondary voltage (3-10 µs), good for 2-stroke & high rpm.
- **Advantages** : no mechanical wear, constant spark energy at all speeds, easier starting, better emission, accurate **timing** (even distributor-less DIS / coil-on-plug).

2.19 Spark Advance & Retard Mechanisms

- Burning needs **time** (≈ 1 ms delay) so spark must be given **before TDC (advance)**. As speed rises, the crank turns more in the same time ⇒ more advance required; at light load mixture is weak & slower ⇒ more advance; at high load, less.



- **Centrifugal advance** (speed) : bob weights on the distributor shaft fly outward against springs → rotate the cam forward relative to shaft ⇒ points open earlier.
- **Vacuum advance** (load) : manifold vacuum on a diaphragm turns the breaker plate against cam rotation ⇒ advance at part load (high vacuum); retarded at full throttle (low vacuum).
- **Retard** : spark is delayed (set manually in old engines) for starting, to avoid kickback & knocking at low speeds. Octane selectors in some engines. Modern ECUs use a **map** with knock-sensor feedback.